# GERMÁN KÜBER

## Senior Blockchain Engineer — Rust, Node.js / TypeScript, MEV & Multi-Chain Infrastructure

Las Palmas, Spain · EU Citizen — authorized to work across the EU/EEA, no visa sponsorship required
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## PROFESSIONAL SUMMARY

Senior backend and blockchain engineer with 15+ years in distributed systems and 4+ years shipping production blockchain infrastructure. Polyglot across **Rust (tokio, async), TypeScript / Node.js, Python / Cython and .NET**, building and operating high-volume on-chain transaction systems across EVM (Ethereum, Arbitrum, Optimism, Base) and non-EVM chains (Solana, StarkNet, Polkadot). At Borderless, operated MEV searcher pipelines processing $50M+ in daily on-chain volume, integrating directly with DeFi protocols — AMMs (Uniswap V2/V3, Curve, Raydium, PumpSwap) and lending markets (Aave) — using **ethers.js, viem, web3.js and @solana/web3.js** alongside Rust-side execution. Deep hands-on knowledge of transaction lifecycles, consensus mechanisms, smart-contract interaction, on-chain data and cryptographic primitives. Microservices in production CI/CD (Kubernetes, AWS/Azure). Active open-source contributor through 80+ international talks and Net-Baires (100K+ members). EU citizen (Spanish nationality) with full authorization to work across the EU/EEA — no visa or sponsorship required — open to self-funded relocation anywhere in the EU.

## CORE EXPERTISE

**MEV Infrastructure:** MEV searcher pipelines, transaction delivery, validator / TPU routing, mempool analysis, sandwich attacks, arbitrage detection, liquidation detection, gas optimization, sub-slot execution. Flashbots stack: MEV-Boost, Suave, MEV-Share, PBS (Proposer-Builder Separation), relays. Jito stack: bundles, BAM (Block Assembly Marketplace), BuilderNet, Jito-Solana, Agave

**Blockchain & Web3:** EVM (Ethereum, Arbitrum, Optimism, Base), Solana, StarkNet, Polkadot SDK / Substrate / FRAME / XCM, parachains, runtime upgrades, smart contracts, account abstraction (ERC-4337), cross-chain flow modeling, DEX integration (Uniswap V2 / V3, Curve, Raydium, PumpSwap), lending markets (Aave), Ethereum clients (Geth reverse-engineering, familiar with Reth, Lighthouse, Prysm)

**Programming Languages & Smart Contracts:** Rust (tokio, async), TypeScript / Node.js, Solidity, Cairo, Python, Cython, Go, C++, JavaScript, C# / .NET

**Web3 Tooling & Libraries:** ethers.js, viem, web3.js, @solana/web3.js, Foundry / Forge, Anchor (Solana), Hardhat

**Distributed Systems & Networking:** Low-latency networking (sub-10ms execution), async runtimes (tokio), concurrency, multi-threaded systems, fault tolerance, high-throughput pipelines, P2P networking, consensus protocols (hands-on with GRANDPA / BABE, familiar with Tendermint and Raft), gRPC, Protocol Buffers, event-driven architectures

**Cryptography & Zero-Knowledge:** ZK proofs (Noir, Barretenberg, Garaga on StarkNet, Cairo), ZK-SNARKs / STARKs, account abstraction (ERC-4337), digital signatures, hashing, Merkle trees, cryptographic primitives applied to MEV, on-chain monitoring, and verifiable computation

**Observability & Tooling:** Prometheus, OpenTelemetry, tracing crate, custom Rust telemetry (50+ real-time production metrics), structured logging, post-mortems, on-call discipline

**AI / LLM Systems:** LLM orchestration (OpenAI, Anthropic, Gemini), LangChain, LangGraph, MCP (Model Context Protocol), RAG pipelines, multi-agent architecture, real-time audio pipelines, multi-provider routing

**Cloud & Infrastructure:** AWS, Azure, Docker, Kubernetes, FastAPI, SignalR, CI / CD pipelines, observability stacks, Linux internals

**Engineering Leadership:** Technical leadership for blockchain and distributed systems teams, hiring and technical interviewing for Rust / blockchain hires, code review and mentoring, technical roadmap, architectural decision records (ADRs), on-call and incident response, open-source contribution practices

## PROFESSIONAL EXPERIENCE

### Independent AI Consultant — Agent Engineering & Enterprise AI Delivery

*Self-employed (autónomo, Spain) — Mar 2026 – Present — Remote, client under NDA*

- Forward-deployed AI consultant embedded with an enterprise engineering organization: work directly with the customer to run discovery with business and technical stakeholders, architect the solution, and own delivery from pilot through production and continuous iteration.
- Design and ship production agentic workflows on LangChain, LangGraph and Microsoft Agent Framework, including multi-agent architectures on Google's A2A (Agent2Agent) protocol with a central capability registry.
- Built knowledge graphs (Cognee, Graphiti) over internal repositories, including specialized code graphs that map microservice architecture so agents can query real system structure when making decisions.
- Own the RAG platform end-to-end, with eval frameworks and regression test suites gating every pipeline change reaching production.

### Lead Software Engineer (Blockchain / MEV)

*Borderless Capital / CTF Capital — Sep 2021 – Mar 2025 — Remote*

- Shipped and operated production MEV searcher infrastructure in Rust (tokio, async) and Python / Cython across EVM (Ethereum, Arbitrum, Optimism) and Solana, owning architecture, code review, mentoring, and performance optimization while remaining hands-on in production code.
- Migrated the system from Python / Cython to Rust and took it to 20x more transactions per second.
- Built Node.js / TypeScript services and tooling that interact with DeFi protocols on-chain — AMMs (Uniswap V2/V3, Curve, Raydium, PumpSwap) and lending markets (Aave) — using ethers.js, viem, web3.js and @solana/web3.js for protocol analysis, on-chain data reads and transaction submission.
- Integrated with the Flashbots stack — MEV-Boost, Suave, MEV-Share, PBS, and relays — for Ethereum MEV execution and bundle submission.
- Integrated with the Jito ecosystem on Solana — bundles, BAM (Block Assembly Marketplace), BuilderNet, Agave, and the Jito-Solana fork — for sub-slot transaction inclusion and validator routing.
- Pipelines processed 10K+ opportunities per day from $50M+ in daily on-chain flow and contributed $100K+ in attributable annual P&L with ultra-low-latency execution under 10ms p99.

- Built custom transaction delivery infrastructure with parallel broadcasting across RPC endpoints and direct validator / TPU routes, achieving sub-slot Solana inclusion (~400ms).
- Designed strategy logic for liquidation detection, multi-hop arbitrage across 5+ DEXs, and cross-chain flow modeling across EVM and Solana.
- Reverse-engineered Geth and Reth to improve transaction simulation and gas estimation, contributing to the REVM community.
- Ran technical due diligence on 5+ projects that ended up getting funded — architecture, risk, and feasibility — acting as the CTO's technical reference on the low-level side.
- Engineered low-latency observability with Prometheus and custom Rust telemetry, instrumenting 50+ real-time metrics across slot timing, validator rotation, and execution outcomes.
- Helped build the blockchain team, which grew from 4 to 10 engineers — code review, mentoring, technical interviewing, and 24/7 incident response with on-call rotation and post-mortems.

**Lead AI Architect & Co-Founder**

*KliverAI (Own Venture, closed Feb 2026) — Mar 2025 – Feb 2026 (full-time) — Remote, Spain*

- Designed a Zero-Knowledge proof system using Noir, Barretenberg, and Garaga on StarkNet for verifiable expert credentials in a decentralized evaluation marketplace.
- Owned technical strategy, architectural decisions, and delivery as the senior-most engineer; led platform development end-to-end on a polyglot stack (Node.js / TypeScript, Python, .NET, Kubernetes, AWS).
- Rewrote the real-time layer from app-embedded SignalR to Azure SignalR Service with Redis-backed state, scaling concurrent users by 500% while holding sub-200ms latency on audio pipelines (OpenAI Realtime API).
- Benchmarked TTS / STT services (ElevenLabs, Deepgram, AssemblyAI, OpenAI Whisper, Play.ht) on latency, cost, and quality — getting latency to 90ms and word accuracy to nearly 95%.
- Built an adaptive simulation engine with multi-agent architecture on LangGraph, including multi-step branching logic and RAG-powered personalization.
- Integrated three LLM providers (OpenAI, Anthropic, Gemini) with dynamic model routing, picking the best model per task to cut inference spend.
- Hired and led a distributed team of up to 3 engineers, conducting technical interviews and final hiring decisions.

**Fractional CTO (Part-time, hands-on)**

*Rocking Product — Apr 2025 – Dec 2025 (part-time, parallel to KliverAI) — Remote*

- Came in as Fractional CTO to an 8-engineer team and restructured it in the first quarter without stepping away from the code; ran technical interviews and hiring decisions.
- Built an image embeddings service (Azure AI Vision and CLIP) to improve product matching across brand comparisons, raising accuracy and lifting the client's revenue by 40%.
- Cut production bugs by 70% through team training and automated processes (testing, CI / CD, code review).
- Optimized LLM usage and cloud infrastructure through model selection, prompt compression, and semantic caching, cutting combined API and cloud spend by 45%.
- Designed a reusable base architecture on LangChain and LangGraph with monitoring via Langfuse and LangSmith; introduced Spec-Driven Development (SDD); proposed and led a full-stack migration from a fragmented JavaScript codebase to .NET.

**Senior Software Architect**

*Paramo Technology — Oct 2020 – Sep 2021 — Remote*

- Owned the end-to-end architecture of a high-traffic online casino (iGaming) platform handling real-time money operations and regulatory reporting, working directly with the CTO.
- Spotted the issues slowing teams down and designed action plans that raised productivity by 40%, measured by features shipped and fewer errors.
- Designed and rolled out event-driven microservices in .NET handling real-time game state, wallet operations, and regulatory reporting; migrated relational databases to MongoDB where it made sense.
- Implemented distributed tracing (OpenTelemetry, Application Insights, Datadog) to catch bugs early and resolve production issues faster; brought in Kubernetes to scale and built automated deploy pipelines.
- Led part of the hiring and designed the technical entry exam; delivered internal training on architecture patterns, testing discipline, and operational best practices.

**Software Architect**

*CloudX (Software Consultancy) — Sep 2017 – Oct 2020 — Remote / Buenos Aires (first employee)*

- Was the company's first employee; led architecture decisions across several enterprise product lines, often working on-site with clients, and helped build part of the team as it grew.
- Designed Angular and React frontends and Node.js / TypeScript and .NET microservices backends using Domain-Driven Design across multiple enterprise product lines.
- Built a distributed system over geospatial data with PostGIS for a UK client; delivered technical training and conference talks to partners and internal teams.

**Software Architect (Angular / .NET)**

*Majestic Hotel Group — Jan 2014 – Aug 2017 — Argentina*

- Designed and shipped full-stack features in Angular and .NET for internal hotel management platforms, first as Team Lead and later as architect; led the migration from AngularJS to Angular 2.
- Redesigned the data model toward an event-oriented architecture, reducing coupling across integrations.

**.NET Web Developer**

*Plus54 — 2011 – 2014 — Buenos Aires (earlier: Acsys 2009–2011, IBM 2008–2009)*

- Early career: built .NET web applications, internal tooling, and mobile components end-to-end; integrated external APIs with unit testing and reliable data exchange patterns.

## KEY ACHIEVEMENTS

**MEV Infrastructure at Borderless** — Rust and Cython pipelines processing $50M+ in daily on-chain volume and contributing $100K+ in attributable annual P&L.

**Polkadot Builder Program Graduate** — Completed Polkadot Blockchain Academy in Lisbon, Portugal (2026).
**Microsoft MVP, Developer Technologies** — 8 consecutive years (2017–present), recognized for technical content and community contributions.
**80+ International Technical Talks** — .NET Conf, DevFest, NetCoreConf, vOpen, Global Azure Bootcamp, and meetups across LATAM and Europe.
**Net-Baires Founder** — Founded and grew Argentina's largest .NET developer community to 100K+ members across 10+ years.
**Microsoft Student Partner** — 5 years (2016–2021).

## EDUCATION & CERTIFICATIONS

Microsoft Certified: Azure Solutions Architect Expert (AZ-305) — Microsoft
Microsoft Certified: Azure Associate (AZ-104 / AZ-204 level) — Microsoft
Polkadot Builder Program — Polkadot Blockchain Academy, Lisbon, Portugal (2026)
Systems Engineer — Universidad Abierta Interamericana (UAI), Buenos Aires, Argentina (2017–2021)
Microsoft MVP, Developer Technologies — Feb 2017 – Present (8 consecutive years)

## OPEN SOURCE & COMMUNITY LEADERSHIP

**Net-Baires (.NET Community)** — Founder (2015–present). Argentina's largest .NET developer community, 100K+ members, 10+ years of monthly meetups and conferences.
**Crypto Para Mortales** — Founder (2023–present). Spanish-language blockchain and crypto education conference.
**Patrones Para Mortales** — Founder (2020–present). Educational content brand on software design patterns and architecture.
**YouTube channel (youtube.com/@germankuber)** — technical deep-dives on Rust, blockchain, AI, and software architecture.

## SPOKEN LANGUAGES

Spanish: Native — English: Full Professional Proficiency — daily use in international remote teams